

Muhammad Assya Dili

Software Engineer

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PROJECTS

SPORTS FIELD SERVICE – PHP, NodeJS, MySQL [Project Link](#)

- Architected a microservices system with 1 API gateway routing to 3 independent backend services (auth, booking, field), enforcing a shared-secret middleware layer so that **100% of internal services are inaccessible except through the gateway**.
- Implemented JWT-based authentication with access/refresh token rotation and a token blacklist for instant revocation; secured **approximately 62% of all API endpoints** with role-based authorization.
- Created granular role-based access control, restricting **about 28% of endpoints** to admin-only operations across booking and field management.
- Built rate limiting at the gateway layer (60 requests/minute per client) to mitigate brute-force and abuse on authentication endpoints.
- Engineered a distributed booking flow with slot-locking via inter-service HTTP calls, ensuring **zero double-booking** through atomic slot status checks before booking confirmation.

ROOM COMFORT MONITORING – Node.js, Laravel, Python (FastAPI), RabbitMQ, MySQL, Docker, Kubernetes [Project Link](#)

- Designed a **9-container microservices system** (gateway, OAuth2 server, Laravel backend, IoT bridge, ML service + worker, MySQL, RabbitMQ) and **orchestrated it on Kubernetes** with Deployments, ConfigMaps, Secrets, and Ingress — **planned for zero-downtime rolling deployments** across all services.
- Built an **event-driven telemetry pipeline** via RabbitMQ that runs **3 ML models per sensor reading** (comfort classification, busy-hour prediction, anomaly detection) and **automatically triggers IoT device actuation (relay/LED)** through a closed-loop callback — no manual intervention required.
- Secured **100% of internal services behind a gateway-only access policy** with **57% of API endpoints restricted to admin-only** operations and **75% requiring authenticated access**.
- Developed a **weighted seat-recommendation engine** (30% position score, 50% isolation score, 20% popularity score) so students receive data-driven seat recommendations to enhance decision-making accuracy.

EXPERIENCE

VETERAN TECH [Project Link](#)

Jakarta Selatan

Members

2026 - present

- Developed high-performance RESTful admin endpoints for *Dignition* (a learning platform) using Hono, Drizzle ORM, and Zod, improving data retrieval efficiency for mentor/mentee relations by **15%**.
- Achieved **100% test coverage** and full OpenAPI compliance across all new endpoints, reducing integration errors during frontend deployment.
- Reduced manual capstone-room management effort by **up to 20%**, delivering meaningful efficiency impact for mentors coordinating cross-Study-Club collaborations.

EDUCATION

UNIVERSITAS PEMBANGUNAN NASIONAL VETERAN JAKARTA

Jakarta Selatan

Undergraduate Student – Computer Science

2024-2028

- GPA: 3.92 / 4.0

- Relevant Coursework: Data Structures and Algorithms, Object Oriented Programming, Service-Oriented Architecture (SOA)
- Clubs: Veteran Tech

ADDITIONAL INFORMATION

- Programming Languages: HTML, CSS, Typescript, Hono, Express, Laravel 13, FastAPI, Docker, Kubernetes, IoT
- Database: PostgreSQL, MySQL, MongoDB
- Tools: Linear, GitHub, Git
- Certifications: Semi-finalist IBCC Ganesha 2026